

PIETERMARITZBURG, South Africa

WMO: 685810

Lat: 29.6278S

Lon: 30.4028E

Elev: 2208

StdP: 13.56

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	40.4	42.6	26.1	21.8	63.2	29.4	25.4	59.4	12.4	63.2	10.5	69.3	0.9	240	0.284

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.0	93.3	70.0	90.1	70.1	87.0	70.1	76.4	85.5	74.9	83.4	73.6	81.6	6.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	137.4	81.3	72.6	131.1	79.4	71.3	125.3	77.8	41.7	85.3	40.1	83.5	38.8	81.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	9.8	8.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			34.4	79.2	3.4	2.3	32.0	80.8	30.0	82.2	28.1	83.5	25.6	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	67.0	73.1	74.1	72.9	68.5	63.8	58.8	58.6	62.6	65.2	66.6	68.5	71.1	(6)
(7)		DBStd	7.59	5.21	5.06	5.29	5.50	4.79	4.20	5.03	5.91	6.89	6.61	6.39	5.78	(7)
(8)		HDD50	7	0	0	0	0	0	1	4	1	1	0	0	0	(8)
(9)		HDD65	823	4	2	6	28	79	190	206	116	80	63	35	14	(9)
(10)		CDD50	6192	717	675	710	556	428	265	270	392	457	513	554	655	(10)
(11)		CDD65	1538	257	257	252	134	42	4	7	42	87	112	140	204	(11)
(12)		CDH74	9812	1537	1467	1436	719	317	88	116	399	712	839	933	1249	(12)
(13)		CDH80	3308	543	496	500	223	62	7	10	117	266	310	341	433	(13)
(14)	Wind	WSAvg	3.0	3.3	3.1	2.8	2.5	2.3	2.3	2.6	3.0	3.3	3.6	3.6	3.5	(14)
(15)	Precipitation	PrecAvg	34.3	5.4	4.6	4.3	1.9	0.8	0.5	0.6	1.0	2.0	3.2	4.1	4.7	(15)
(16)		PrecMax	50.6	11.8	12.4	11.7	6.1	3.9	2.4	4.3	3.4	17.5	7.0	10.3	11.5	(16)
(17)		PrecMin	23.0	1.9	1.0	0.8	0.2	0.0	0.0	0.0	0.0	0.2	0.9	1.4	1.3	(17)
(18)		PrecStd	6.6	2.3	2.6	2.6	1.3	0.8	0.6	0.9	0.9	2.9	1.5	1.9	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.0	95.1	94.0	92.0	86.2	80.5	81.2	88.6	94.5	95.0	95.3	94.3	(19)
(20)			MCWB	71.9	72.2	70.8	67.4	60.9	57.2	55.7	59.6	63.1	66.4	68.9	71.7	(20)
(21)		2%	DB	90.2	89.9	89.2	85.7	81.3	76.2	77.4	82.9	87.3	88.3	89.0	88.8	(21)
(22)			MCWB	73.3	73.3	71.8	67.0	61.3	56.0	55.5	58.7	62.8	67.0	70.4	72.7	(22)
(23)		5%	DB	85.8	85.7	85.4	81.7	77.4	72.7	73.4	78.2	81.8	83.3	84.0	84.8	(23)
(24)			MCWB	73.1	73.3	71.6	66.4	60.8	55.1	55.3	58.1	62.3	66.8	69.9	72.5	(24)
(25)		10%	DB	82.0	82.3	81.9	77.6	73.7	69.1	69.1	73.6	77.0	78.3	79.3	81.1	(25)
(26)			MCWB	72.3	72.5	70.8	65.8	60.4	54.6	54.3	57.9	61.9	65.5	68.5	71.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	78.2	77.0	71.6	66.6	61.6	61.1	65.6	68.9	72.9	76.0	77.9	(27)
(28)			MCDWB	86.9	87.3	86.0	82.9	77.1	70.4	74.1	75.7	82.3	85.4	87.3	87.4	(28)
(29)		2%	WB	75.6	76.0	74.6	69.8	64.7	59.7	58.8	63.1	66.3	69.9	73.1	75.6	(29)
(30)			MCDWB	84.3	84.4	82.9	79.3	73.5	67.3	69.7	73.2	78.9	80.7	82.8	83.9	(30)
(31)		5%	WB	74.1	74.4	72.9	68.3	63.2	57.9	57.2	61.1	64.6	68.1	71.1	73.6	(31)
(32)			MCDWB	82.3	82.1	81.0	76.2	71.1	66.7	66.5	71.5	75.3	78.2	79.3	81.4	(32)
(33)		10%	WB	72.6	73.1	71.5	66.9	61.9	56.3	55.8	59.4	63.0	66.5	69.4	71.9	(33)
(34)			MCDWB	80.2	79.9	79.2	74.1	69.6	65.6	65.1	69.7	72.4	74.9	76.9	78.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.7	16.0	17.1	19.0	23.1	25.1	24.6	22.8	20.5	18.1	16.2	15.6	(35)
(36)			MCDBR	23.5	22.1	23.7	25.8	28.5	30.2	30.5	31.3	30.7	29.2	26.3	23.4	(36)
(37)			MCWBR	9.0	8.2	8.5	8.6	10.5	12.1	12.8	10.8	10.5	10.8	10.1	10.1	(37)
(38)		5% WB	MCDWB	20.2	19.2	20.2	20.7	22.4	23.8	25.3	23.3	24.6	23.1	21.0	20.7	(38)
(39)			MCWBR	9.1	8.6	8.8	8.3	10.0	11.7	12.8	9.9	10.4	10.3	10.2	10.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.422	0.416	0.404	0.369	0.341	0.323	0.333	0.394	0.443	0.459	0.432	0.426	(40)	
(41)		taud	2.314	2.332	2.346	2.400	2.440	2.465	2.426	2.258	2.144	2.148	2.254	2.290	(41)	
(42)		Ebn at Noon	292	288	281	275	268	266	267	262	264	272	287	292	(42)	
(43)		Edh at Noon	44	42	39	33	29	27	29	38	47	50	46	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1815	1792	1593	1328	1156	1037	1113	1291	1485	1571	1709	1800	(44)	
(45)		RadStd	132	143	82	86	64	54	65	78	93	152	147	127	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page